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STATISTICS

Session 1: Introduction to Statistics & Data

What is Statistics?

When people hear the word “statistics”, they tend to think about:

- Numbers
- Formulas
- Calculations
- Tables
- Graphs
- Averages
- Percentages

However, statistics is <u>not</u> simply the calculation of numbers, and the art of making graphs look pretty.



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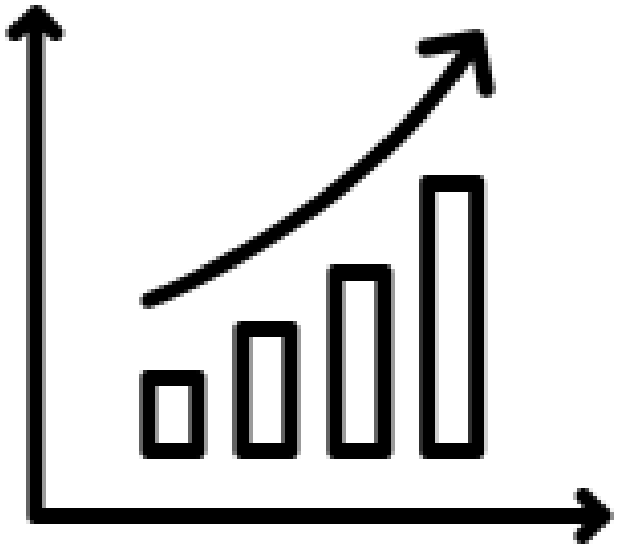
What is Statistics?

Statistics is the process of using data to:

- Answer questions
- Learn about the world
- Make decisions when we do not have complete information.

For example, Statistics can help us to answer questions such as:

1. Is a new advertising campaign increasing sales?
2. Which customers are most likely to stop using a service?
3. Does a new medical treatment work?
4. Is a change in customer satisfaction meaningful, or could it simply be random variation?
5. What is likely to happen in the future?



Why study Statistics?

- For students studying quantitative courses (e.g. mathematical sciences, economics and econometrics, etc.), they will naturally take several statistics units during their studies.
- Students in other courses (e.g. Medicine, Psychology, Sport Science, Engineering, IT, Data Science, etc.) will usually take at least one statistics unit during their degree.
- The question becomes, why do all students need statistical training at some point in their degree?
- This is because statistics teaches you not just how to calculate but how to **think**: how to challenge claims, test assumptions, and make sense of messy real-world data.



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Why study Statistics?

- On 28 January 1986, the space shuttle *Challenger* exploded only about 70s after launch. This disaster was viewed by millions of people on live television.
- Based upon previous tests:
  - Engineers had warned that cold temperatures could affect the rocket's O-ring seals.
  - The analysis focused on flights with damage, leaving out flights with zero damage. That made the relationship between temperature and damage look unclear.
- To assess the risk, the right statistical model would use all flights to estimate how the probability of damage changes with temperature.
- NASA did not use the right statistical model and went ahead with the launch anyway.



SPECIAL REPORT ON SHUTTLE DISASTER / 9 PAGES

San Francisco Chronicle

The Largest Daily Circulation in Northern California

122nd Year No. 11 ★★★★★★ WEDNESDAY, JANUARY 29, 1986 777-1111 25 CENTS

WHAT WENT WRONG

7 Killed In Shuttle Explosion

By Boyce Rensberger, Washington Post

Cape Canaveral

The space shuttle Challenger, carrying a crew that included schoolteacher Christa McAuliffe, exploded in a burst of flames 74 seconds after liftoff yesterday, killing all seven aboard and stunning a world made witness to the event by television.

The unexplained explosion occurred without warning as the flight seemed to be proceeding flawlessly at about 2000 feet per second, 10 miles above Earth and eight miles downrange from Cape Canaveral.

The spacecraft disintegrated into bits of debris that rained into the Atlantic Ocean. Those aboard, still strapped into their seats, had no means of escape.

It was the worst accident in the history of space exploration and the first time anyone has been killed during an American space flight.

The tragedy occurred 19 years and one day after U.S. astronauts Virgil (Gus) Grissom, Roger Chaffee

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Smoke plumes trailed the solid fuel rockets (upward, right) as they continued skyward after the liquid fuel tank and shuttle exploded

2 Areas on Fuel Tanks Suspected

By Thomas H. Maugh II, Los Angeles Times

Cape Canaveral

A little more than a minute after the space shuttle Challenger lifted off from pad 39B at the Kennedy Space Center yesterday morning a thin tongue of flame appeared between the left booster rocket and the shuttle's main fuel tank.

The flame was not seen by observers on the ground, but slow motion videotapes of the launch show it clearly.

Less than a second later, nearly 200,000 pounds of liquid hydrogen in the main fuel tank exploded in a massive fireball that enveloped the shuttle orbiter and sent both solid rocket boosters flying off in near opposite directions. The explosion destroyed the orbiter.

Although there is little doubt about what happened, exactly how it happened remained a major mystery. Yesterday, however, most of the speculation centered on two possibilities.

One is that a defect in the solid rocket booster caused exhaust

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THE SHUTTLE CHALLENGER'S FATEFUL LIFTOFF

8:39:12 TV tracking camera spots fireball near bottom of Challenger main fuel tank as the spaceship is 10.4 miles high and 8.0 miles from its Florida coast launching site. The shuttle is traveling at 1977 miles per hour.

8:39:13 Challenger explodes, spewing burning pieces like a fireworks display.

8:39:09 Last transmission from Challenger responding to command from Mission Control to increase thrustle: "Challenger. Go at throttle up."

8:38:52 Shuttle is traveling at 1539 miles per hour; altitude is 4.9 miles; distance from shore is 3.4 miles. Mission Control spokesman reports "Engines are throttling up. Three engines are at 104 percent."

8:38:07 Spacecraft begins its normal flight pattern that puts the shuttle on its back in a pre-planned maneuver.

8:38:00 a.m. (PST) Challenger lifts off from launch pad, all systems appear normal.

Tracking equipment reports debris hit the water in an area approximately located at 28.64° North, 80.28° West; the water in the area is 75 to 100 feet deep.

Launch pad 39 B

Main fuel tank

Solid fuel boosters

Serious Doubts Raised Over Future Space Flights

By David Perlman, Science Editor

The Challenger disaster yesterday forced a long delay in America's manned space program and fueled new controversy over its future.

This year was to have been one of the greatest and most adventurous for American space exploration since the space age began nearly 30 years ago, but now the adventure is on hold.

The speedup of the shuttle program's crowded schedule — strongly prodded by the White House — must await the results of NASA specialists who are probing the cause of the explosion that destroyed the Challenger and killed all seven aboard.

Another and even more ambitious effort, a program to put a huge space station in orbit by 1993, may be in far greater jeopardy, experts said.

Until yesterday, 15 space shuttle missions were scheduled for this year, including three to send dramatic scientific payloads into orbit and deep space. At least three secret military flights also were planned from the new shuttle spaceport at California's Vandenberg Air Force Base.

NASA's annual budget, running at nearly $8 billion, has more than $1.5 million earmarked for "space transportation" — the shuttle program with its high-technology development, expensive launches and corps of astronauts.

With the number of shuttles now reduced to three, the first has been grounded by President Reagan until the cause of the catastrophe has been pinpointed. That may take many months, experts say —

President Postpones State of Union Talk

Washington

Saying "today is a day for mourning and remembering," President Reagan postponed the State of the Union address because of the space shuttle explosion.

In brief, nationally televised remarks from the Oval Office late yesterday afternoon, Reagan promised "to continue our quest in space."

"There will be more shuttle flights and more shuttle crews

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■ An unmanned probe called Galileo, which is to orbit the planet Jupiter and parachute instruments into its atmosphere.

■ The 10-ton Hubble Space Telescope, whose 94-inch mirror should enable astronomers to peer out to the very edges of the universe billions of light-years distant.

■ The European-built Ulysses space probe that will orbit the sun from pole to pole to measure solar

and, yes, more volunteers, more civilians, more teachers in space," he said. "This is truly a national loss. We mourn seven heroes. We mourn their loss as a nation together."

Concerned that schoolchildren across the nation had watched the explosion on television and had witnessed the death of Christa McAuliffe, the teacher from Concord, N.H., who was to have been the first ordinary citi-

particles and magnetic fields never before observed.

Postponing the Galileo mission, which was due to be launched in May, is a particular blow to scientists because the trajectory to Jupiter is only possible during a 30-day period in which the planet's orbit is in the right position to be reached from Earth.

It now appears the mission will

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INSIDE

Actress Lilli Palmer Dies at 71

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Why study Statistics?

- Not everyone here is going to work at NASA or be in a role where your decisions could cost lives.
- However, every one of you will be expected to make decisions using data, whether you're:
  - managing a budget,
  - launching a product,
  - analyzing customer feedback, or
  - pitching a new strategy to stakeholders.

- Inaccurate assumptions, poor interpretation of trends, or blindly trusting a single number without questioning where it came from, these mistakes won't make the news like the Challenger disaster did, but they can cost your company money, credibility, or opportunity.

- And more importantly, they can damage your own confidence and reputation as a professional.

Data and Data Tables

- Modern statistics focuses on using data and analytics to answer questions, understand uncertainty and make better decisions.

- In particular, modern statistics has a large focus on making decisions with data.
  - What data do we need?
  - How to collect the data?
  - Which methods are appropriate for analysing data?
  - What conclusions can we make with the data?



Data and Data Tables

Data are recorded pieces of information about people, objects, events or processes.

- A university might record data about its students, including their degree, age, attendance and assessment results.
- A business might record information about each sale, such as the product purchased, the price paid, the store location and the date of purchase.
- A hospital might record a patient's symptoms, test results, diagnosis and treatment outcomes.



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Data and Data Tables

- The people, objects or events we collect information about are often called cases, observations or units.
- The characteristics recorded for each case are called variables
- When we analyse data, we need to begin by having it in a tabular format
- A good data table will have the information organised where each row represents one case, and each column is one variable.

Each column is a variable

Student	Degree	Age	Attendance	Assessment Score
	Arts	22	91%	90
	Science	19	85%	92
	Business	20	92%	89

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Where does data come from?

PRIMARY DATA

- The researcher is directly involved in collecting the data. For example:
  - a researcher might conduct a survey to understand people's attitudes towards public transport.
  - a researcher might run an experiment to compare the effectiveness of two medications.

SECONDARY DATA

- Other data already exist because they were originally collected for another purpose. For example:
  - A researcher might use government census data,
  - hospital records,
  - company sales data or
  - information from an existing database.

Where does data come from?

Common sources of data includes:

Surveys and Questionnaires	People report their opinions, behaviours or characteristics
Experiments	Researchers deliberately change something and observe the outcome
Observational studies	Researchers observe what occurs without directly intervening
Administrative records	Such as enrolment records, tax records and medical records
Business transactions	Including purchases, payments and customer interactions
Sensors and devices	Such as fitness trackers, weather stations, satellites and GPS devices
Digital activity	Including website visits, online searches and interactions with apps or social media

The source of the data matters because data are not simply neutral facts that appear on their own. Someone has decided what to measure, how to measure it, whom to include and when to collect it.

What do we do with data?

- Before we do any analysis, we need to clean and organise our data. Real datasets may contain missing values, incorrect entries, inconsistent labels or duplicated observations.
- The process of cleaning real data is time-consuming, but worth it, because the real fun happens after that.
- With clean data, we then perform exploratory data analysis (EDA). Tables, graphs and summary statistics help us understand what has been observed.
- But the true strength of statistics is to make conclusions beyond the available data. When you see research articles published in journals, or breaking news stories on television/online, these are usually based upon statistical inference.

Business Analytics

Overview | Sales | Customers | Products | Marketing

Jan 1, 2024 – Dec 31, 2024 | All Regions

- Dashboard
- Analytics
- Reports
- Segments
- Targets
- Settings

Revenue
$2,480,000 ↑ 12%

Profit
$620,000 ↑ 18%

Customers
24,893 ↑ 11%

Growth
15.4% ↑ 3.2%

Monthly Revenue



Sales by Product



Customers by Region



Sales by Channel



Customer Distribution



Revenue vs. Profit



Active Users



Sales Funnel



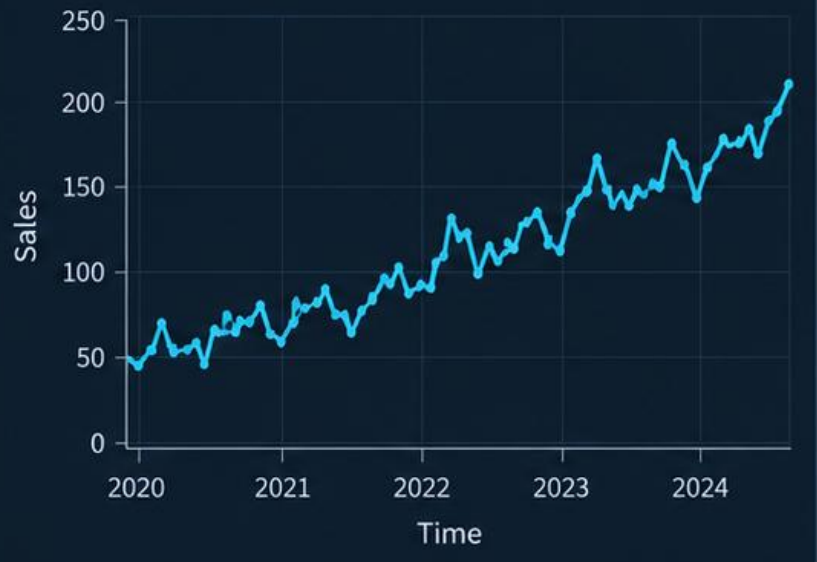
Top Products

#	Product	Revenue	Units	Change
1	Alpha	$420,000	3,200	↑ 12%
2	Beta	$310,000	2,480	↑ 8%
3	Gamma	$268,000	1,980	↑ 15%
4	Delta	$184,000	1,420	↓ 6%
5	Epsilon	$142,000	1,120	↑ 11%

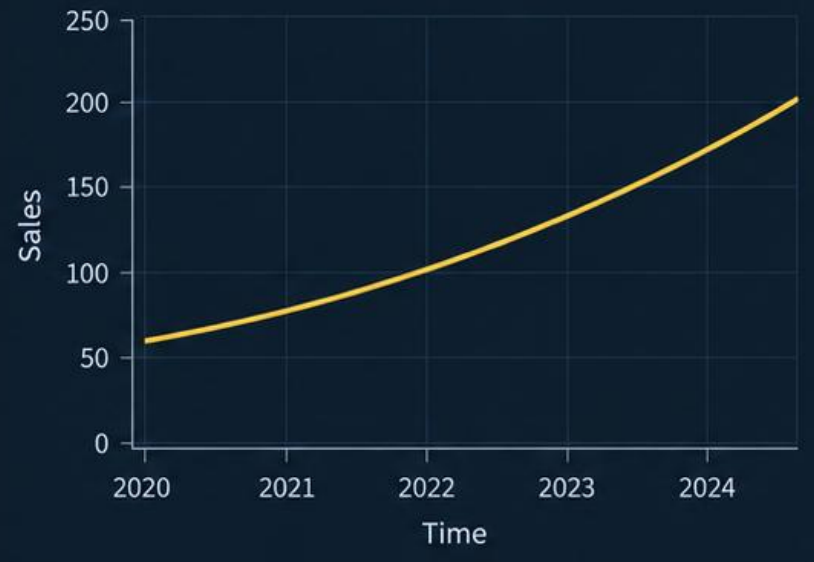
Activity Heat Map



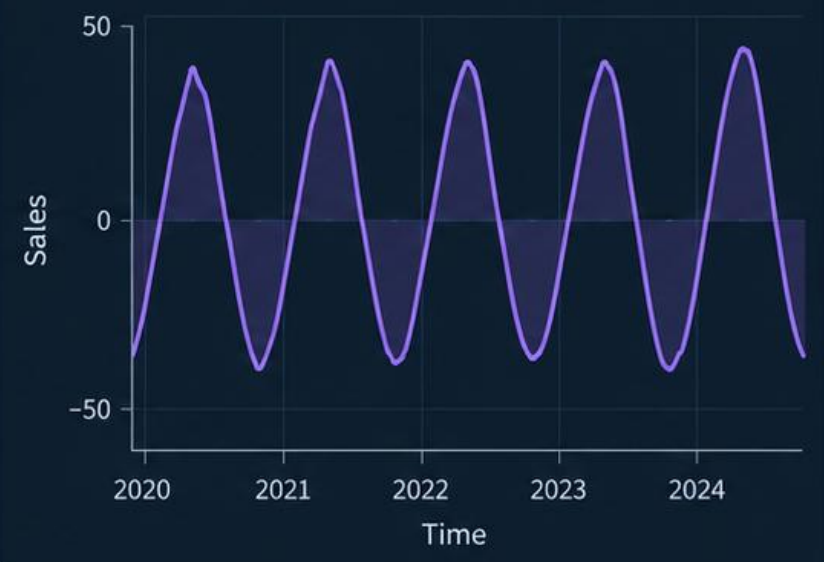
Historical Sales



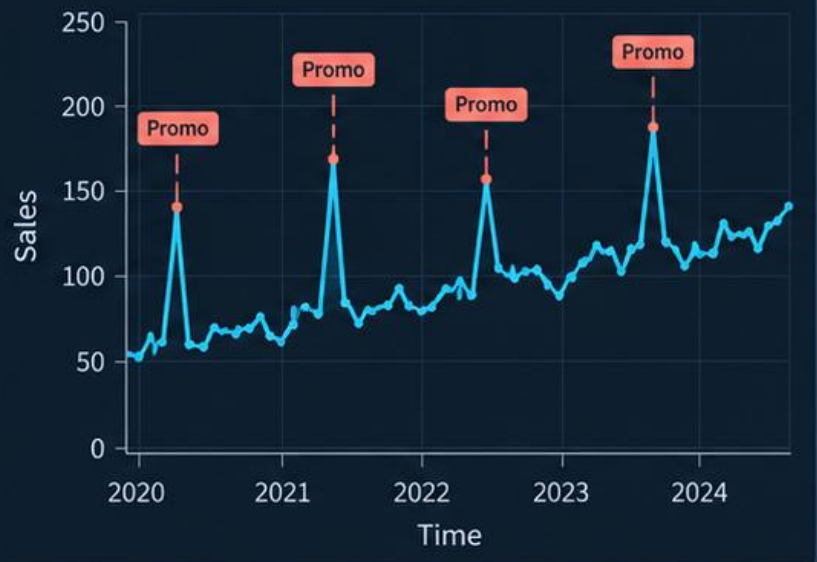
Long-term Trend



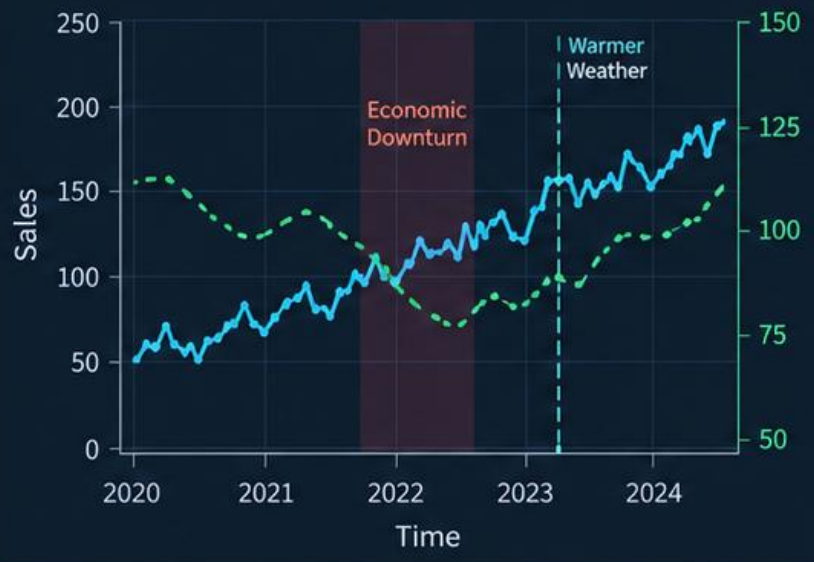
Seasonality



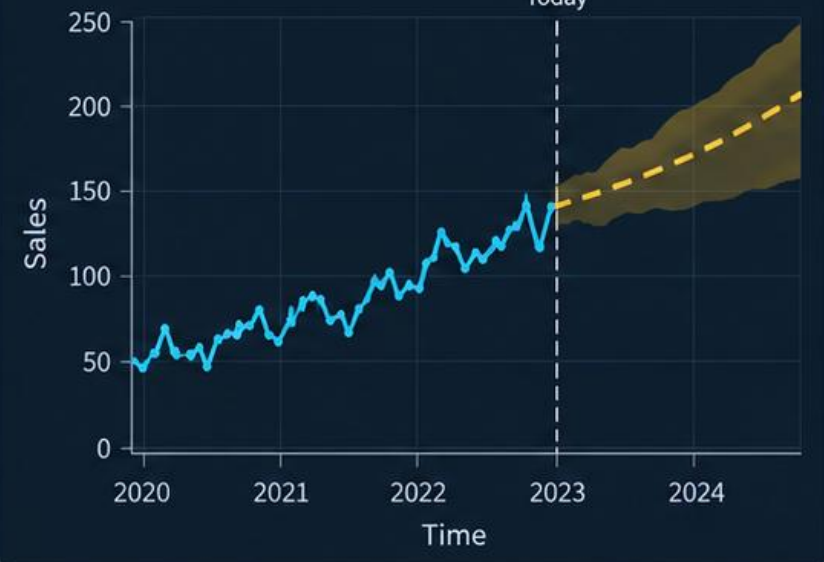
Promotion Effect



External Effects



Sales Forecast



Variables

- A variable is a characteristic or measurement that can take different values across observations.

- The observations might be people, businesses, products, countries or points in time. For example, in a dataset containing information about students, each row might represent one student and each column might represent a variable.

Student	Study time (hours)	Test score
1	2.0	58
2	4.5	76
3	6.0	84
4	3.0	65

Variables

- The amount of studying and the test score can vary from one student to another. One student might study for two hours, while another might study for six hours. Their test scores may also differ.

- Variables can also change for the same person over time. A student might study for five hours for one test but only two hours for another. The same student may obtain different scores across different assessments.

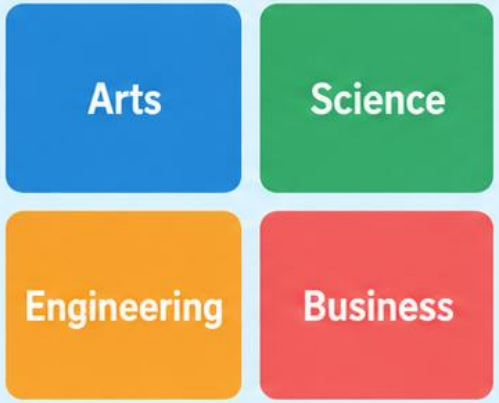
- Therefore, variation can occur:
  - between people, because different students may have different values;
  - within a person, because the same student's value may change over time.

Levels of Measurement

NOMINAL

Categories, no order

Example: course of study



ORDINAL

Ordered categories

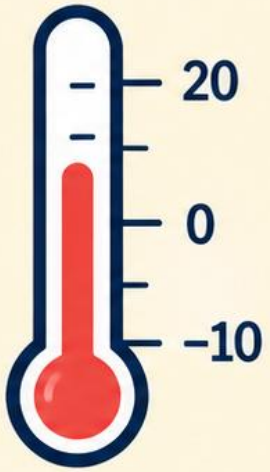
Example: low, medium, high



INTERVAL

Equal intervals, no true zero

Example: temperature (°C)



RATIO

Equal intervals, true zero

Example: study time (hours)



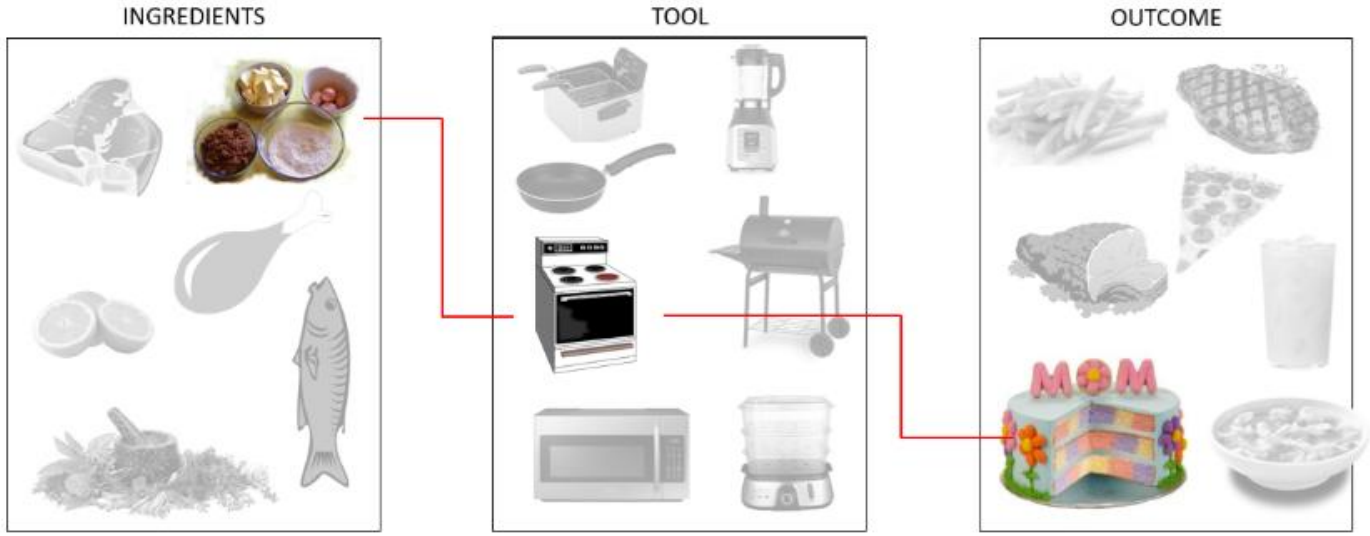
A true zero makes "twice as much" meaningful.

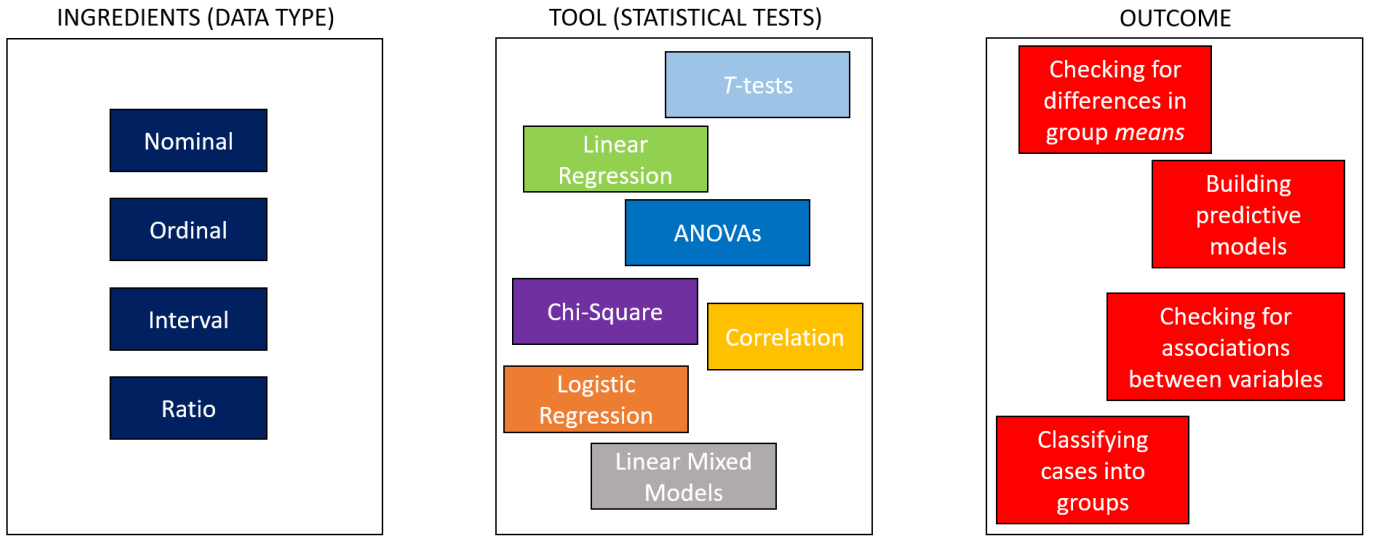
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Statistics is a lot like cooking!

- When cooking a dish, you need to use the right ingredients with the right tools, to obtain the desired outcome! The same principal applies in Statistics as well.

- If we use the wrong statistical tool for the data at hand, our results may be misleading or invalid, just like trying to bake a cake in a deep fryer.

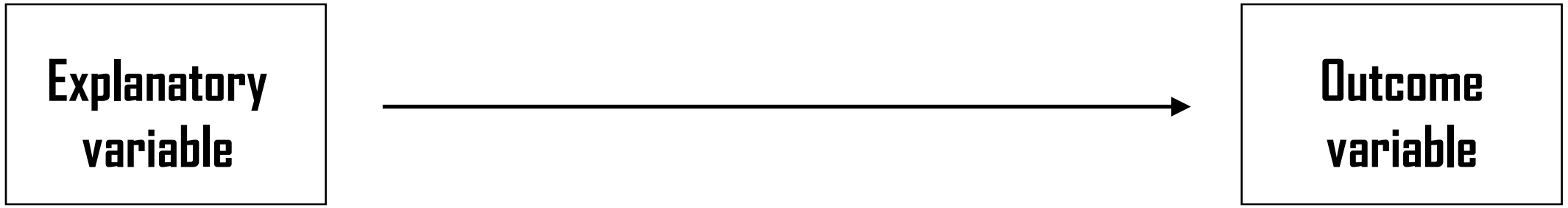




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Explanatory and outcome variables

- When investigating a relationship, variables are often described according to the role they play in the research question.
  - An **outcome variable** is the variable we want to understand, explain or predict. It represents the result of interest.
  - An **explanatory variable** is a variable that may help explain or predict variation in the outcome variable.



Explanatory and outcome variables

Explanatory variable

- An explanatory variable is a variable that might explain, influence or predict changes in another variable.
- It represents the possible reason why outcomes differ. Depending on the context, it may also be called a predictor, independent variable, input or risk factor.
- An explanatory variable can be:
  - Categorical, such as treatment group, teaching method or employment type.
  - Numeric, such as age, temperature, advertising expenditure or hours of sleep.

Outcome variable

- An outcome variable is the variable we want to understand, explain or predict.
- It records the result or response of interest. Depending on the context, it may also be called a response, dependent variable or target.
- An outcome variable can also be:
  - Categorical, such as whether a patient recovered, whether a customer made a purchase
  - Numeric, such as recovery time, sales revenue, examination score or stress level.

Explanatory and outcome variables

